BRAC UNIVERSITY
Department of Computer Science and Engineering

Examination: Surprise Test -2
Duration: 10 minutes

Semester: Spring 2025
Full Marks: 05

CSE 340: Computer Architecture

Name: Solution	ID:	Section: 1

1. Given the contents of register X_{19}

0000 0000 0000 0000 0000 0000 0000 0000 1011 0110 1101 1110 0011 1010 1011 0100

28 25 20 12 8 7 5

For further analysis, we need to extract only the bits at the 5th, 7th, 8th, 12th, 20th, 25th and 28th positions.

a) Construct a mask to do so.
Answer: Mask =

0000

0000 ...... 0000 0001 0010 0001 0000 0001 0001 1010

$\underbrace{\text{0000 ...... 0000}}_{\text{32 bit}}$ 28 25 20 12 8 7 5

b) Write the name of the instruction that can be used to extract (copy) these bits using the prepared mask.
Answer: "And"

2. Construct the equivalent RISC-V code of the following high-level code.

```
if (a > b or c <= d):
    save[i] = 5
else:
    b = 10
```

any of the conditions

The base address of a halfword array save is in register X_{19}. Also consider a, b, c, d and i are in registers X_{20}, X_{21}, X_{22}, X_{23}, X_{24} respectively.

Solution:

Blt x_{21}, x_{20}, If

BGE x_{23}, x_{22}, If

BEQ x_0, x_0, Else

$a > b = b < a$

x_{21} x_{20}

$c <= d = d >= c$

x_{23} x_{22}

If:

Addi x_6, x_0, 5 ; $x_6 = 5$

Slli x_5, x_{24}, 1 ; x_5 = Offset = $i \times 2^1$

Add x_5, x_5, x_{10} ; x_5 = Base + Offset

SH x_6, 0(x_5) ; Save[i] = 5

Beq x_0, x_0 , Exit

Else:

Addi x_{21}, x_0, 10

Exit: